.9 The resistance to exposure to light when tested in accordance with ISO 105-BO2:1988 and the contrast between the unexposed and exposed samples should not be less than class 5.

.10 The resistance to wet and dry rubb ing when tested in accordance with ISO 105-X12:1995 and should not be less than class 3.

.11 The resistance to seawater should not be less than class 4 in accordance

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SOURCE: MSC.200(80).pdf
.10 The resistance to wet and dry rubb ing when tested in accordance with ISO 105-X12:1995 and should not be less than class 3.

.11 The resistance to seawater should not be less than class 4 in accordance with ISO 105 EO2:1978 and the change in colour of the specimen should not be less than class 4.

2.10.4.3 Operating head load test

The operating head load test should be carried out using tw o lifejackets one lifejacket to be conditioned at -30°C for 8 h and the other at +65 °C for 8 h. After mounting on the manikin or the test form the lifejackets should be inflated, and a steady force of 220 10 N applied to the operating head as near as possible to the point where it enters the buoyancy chamber. This load should be maintained for 5 min during which the direction and angle in which it is applied should be continuo usly varied. On completion of the test the lifejacket should remain intact and should hold its pressure for 30 min.

2.10.4.4 Pressure test

2.10.4.4.1 Overpressure test: The inflatable buoyancy ch ambers should be capable of withstanding an internal over pressure at ambient temperature. A lifejacket should be

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SOURCE: MSC.200(80).pdf
2.10.4.4 Pressure test

2.10.4.4.1 Overpressure test: The inflatable buoyancy ch ambers should be capable of withstanding an internal over pressure at ambient temperature. A lifejacket should be inflated using the manual method of inflation, after inflation the reli ef valves should be disabled and a fully charged gas cylinder according to the manufacturers recommendation

should be fitted to the same inflation device and fired. The lifejacket should remain intact

and should hold its pressure for 30 min. The lifejackets should show no signs of damage such as cracking, swelling or changes of mechanical qualities and that there has been no significant damage to the lifejacket inflation component. All fully charged gas cylinders used in this test should be sized according to the markings on lifejacket.

2.10.4.4.2 Relief valve test: With one buoyancy chamber inflated, the operating head

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SOURCE: MSC.200(80).pdf
used in this test should be sized according to the markings on lifejacket.

2.10.4.4.2 Relief valve test: With one buoyancy chamber inflated, the operating head on the opposite buoyancy chamber should be fired manually, using a fully charged gas cylinder according to the manuf acturers recommendations. The operation of the relief valves should be noted to ensure that the excess pressure is relieved. The lifejacket should remain intact and should hold its pressure for 30 min. The lifejackets should no signs of damage such as cracking, swelling or changes of mechanical qualities and that there has been no significant damage to the lifejacket inflation component.

2.10.4.4.3 Air retention test: One inflation chamber of a lifejacket is filled with air until air escapes from the over-pressure valve or, if the lifejacket does no have an over-pressure valve, until its de sign pressure, as stated in th e plans and specifications, is reached. After 12 h the drop in pressure should not be greater than 10%. This test is then repeated as many times as necessary to test a different chamber until each chamber has been tested in this manner.

2.10.4.5 Compression test

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SOURCE: MSC.200(80).pdf
reached. After 12 h the drop in pressure should not be greater than 10%. This test is then repeated as many times as necessary to test a different chamber until each chamber has been tested in this manner.

2.10.4.5 Compression test

The inflatable lifejacket, packed in the normal manner should be laid on a table. A bag containing 75 kg of sand and having a base of 320 mm diameter s hould be lowered onto the lifejacket from a height of 150 mm in a time of 1 second. This should be repeated ten times, after which the bag should remain on the jacket for not less than 3 h. The lifejacket should be inflated by immersion into water and should inflate fully, the jacket to be inspected to ensure that no sw elling or change of mechanical properties has occurred, the jacket should be checked for leaks.

2.10.4.6 Test of metallic components

2.10.4.6.1 Metal parts and components of a lifej acket should be corrosion-resistant to seawater and should be tested in accordance with ISO 922 7:1990 for a period of 96 h. The metal components should be inspected and should not be signific antly affected by

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SOURCE: MSC.200(80).pdf
seawater and should be tested in accordance with ISO 922 7:1990 for a period of 96 h. The metal components should be inspected and should not be signific antly affected by corrosion, or affect any other part of the lifejacket and should not impair the performance of the lifejacket.

2.10.4.6.2 Metal components should not affect a magnetic compass of a type used in small boats by more than 1°, when placed at a distance of 500 mm from it.

2.10.4.7 Inadvertent inflation test

2.10.4.7.1 The resistance of an automatic inflation device to inadvertent operation should be assessed by exposing the entire lifejacket to sprays of water for fixed period. The lifejacket should be fitted correctly to a free-standing manikin of adult size, with a minimum shoulder height of 1,500 mm (see figure 5), or alternatively to an appropriately sized form as shown in figure 2. The lifejacket should be deployed in the mode in which it is worn ready for use but not deployed as used in the water (i.e. if it is equipped with a cover which is normally worn closed, then the cover should be closed for the test). Two


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SOURCE: MSC.200(80).pdf
it is worn ready for use but not deployed as used in the water (i.e. if it is equipped with a cover which is normally worn closed, then the cover should be closed for the test). Two

sprays should be installed so as to spray fr esh water onto the lifejacket, as shown in the diagram. One should be positioned 500 mm above the highest point of the lifejacket, and at an angle of 15 ° from the vertical centre line of the manikin and the bottom line of the lifejacket. The other nozzle should be installed horizontally at a distance of 500 mm from the bottom line of the lifejacket , and points directly at th e lifejacket. These nozzles should have a spray cone of 30 °, each orifice being 1.5 0.1 mm in diameter, and the total area of the orifice should be 50 5 mm, the orifice being evenly spread over the spray nozzle area.

2.10.4.7.2 The air temperature should be 20 °C, and water should be supplied to the

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SOURCE: MSC.200(80).pdf
total area of the orifice should be 50 5 mm, the orifice being evenly spread over the spray nozzle area.

2.10.4.7.2 The air temperature should be 20 °C, and water should be supplied to the sprays at a pressure of 0.3 kPa - 0.4 kPa, a flow of 600 l/h, and a temperature of 18° C

to 20°C.

2.10.4.7.3 The sprays should be turned on, and the lifejacket exposed to the following series of test to access the ability of the jacket to resist inadvertent inflation:

.1 5 min with the high spray on the front of the lifejacket;

.2 5 min with the high spray on th e left side of the lifejacket;

.3 5 min with the high spray on the back of the lifejacket; and

.4 5 min with the high spray on th e right side of the lifejacket.

During exposures .1, .2 and .4, the horizontal spray should be applied for 10 periods of 3 s each to the front, left or right sides (but not back) as with the high spray.

Figure 5 Test set-up for test of automatic inflation system

2.10.4.7.4 After completing the above test, the lifejacket should be removed from the manikin and immersed in water to verify that the auto-inflation system functions.”
RESOLUTION MSC.200(80)

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